



ORAL PITCH PRESENTATION

Speeding up 6-DoF Grasp Sampling with Quality-Diversity

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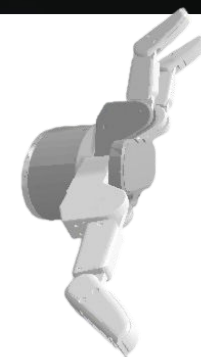


How does a robot grasp an object?

Grasp = select a special configuration of the robot end effector



6-DoF grasp sample

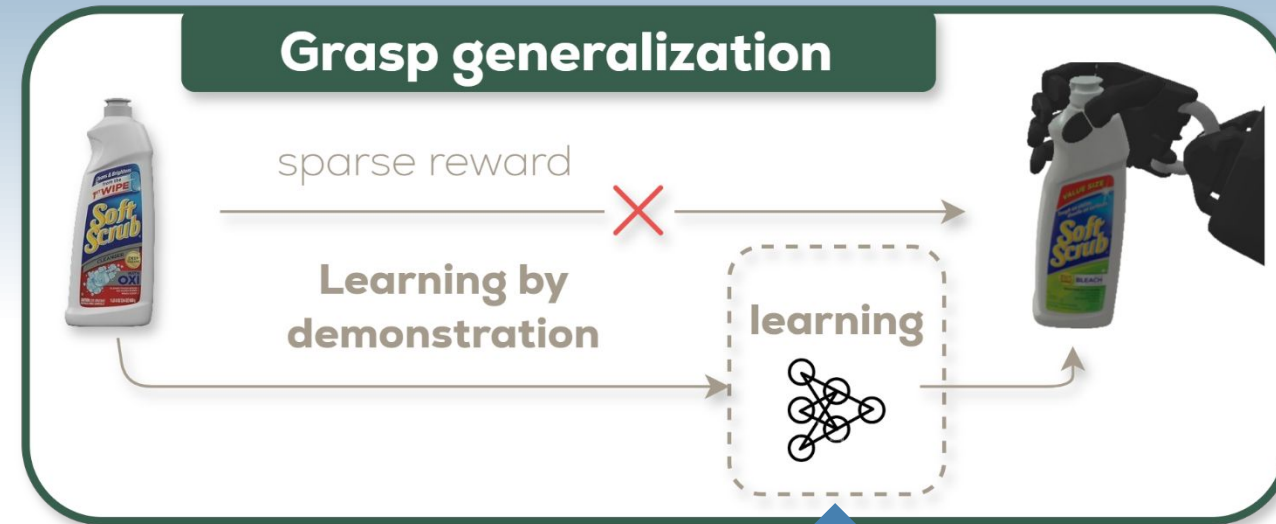


robot end-effector

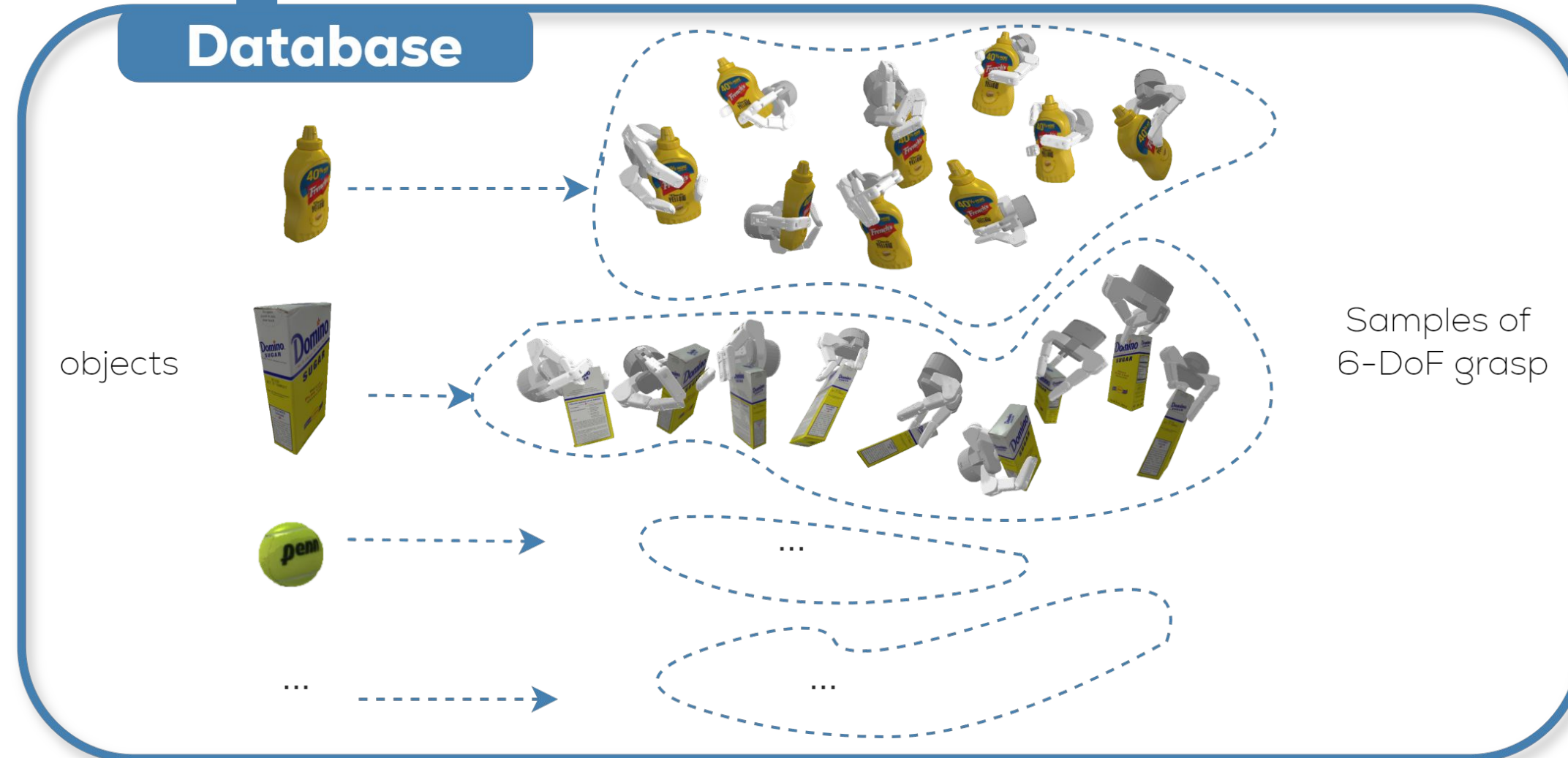


object

Learning to grasp in robotics

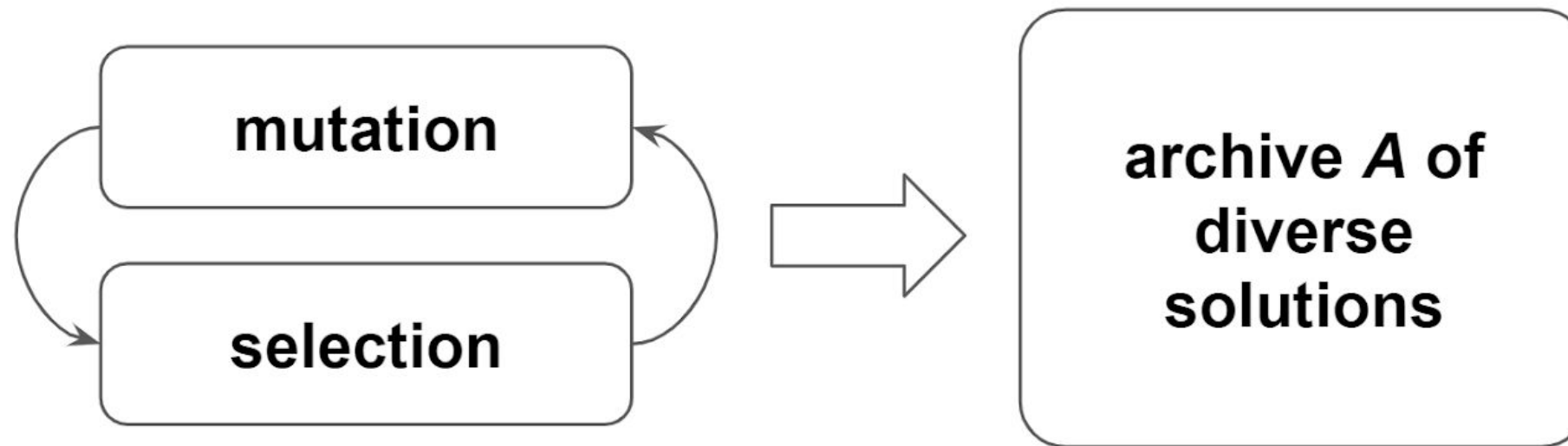


Teach a robot how to grasp any object



Quality-Diversity Methods

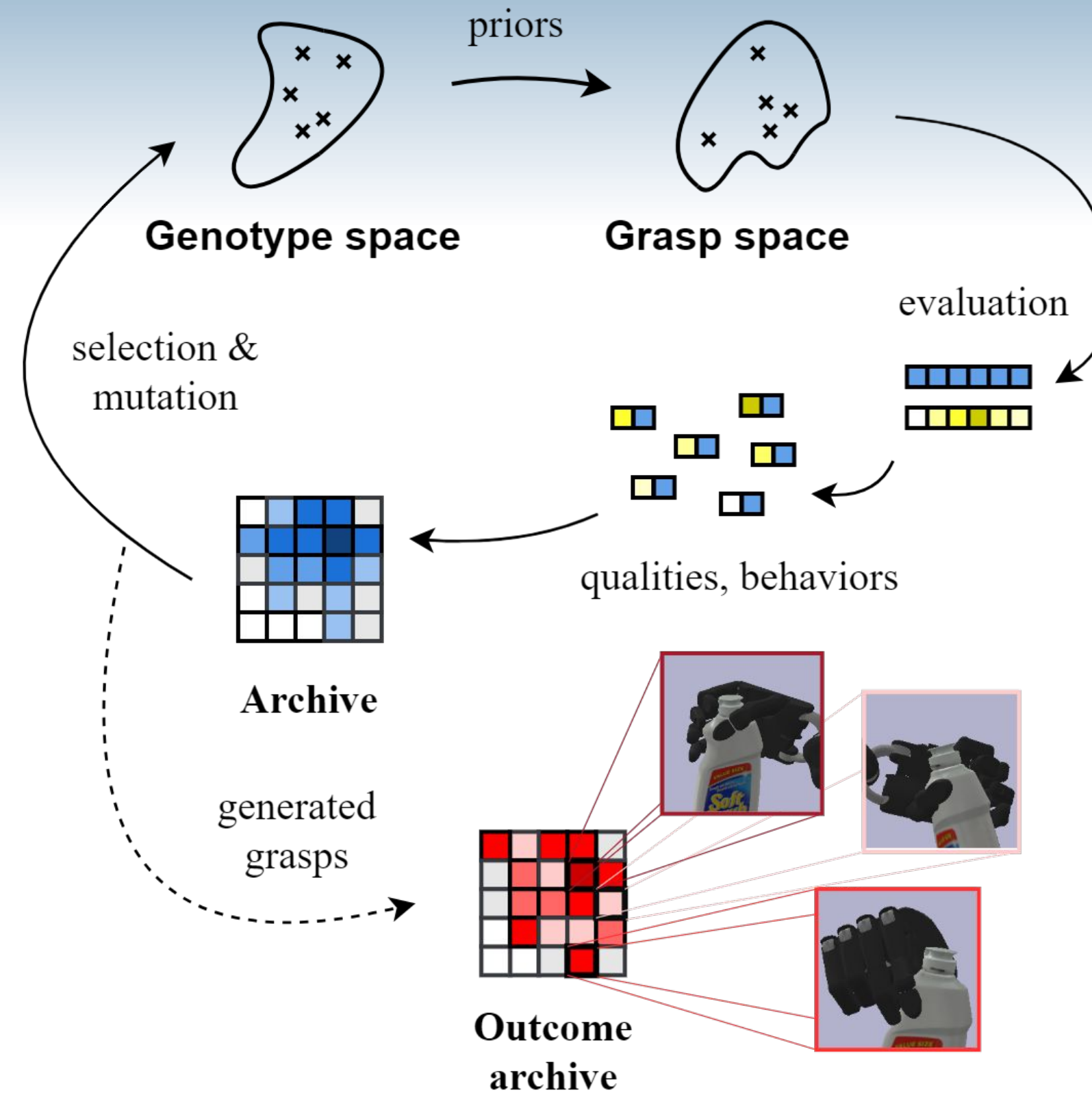
Evolutionary algorithms



$$\begin{cases} \forall b \in \mathcal{B}_{reach}, \exists \theta \in A, d_{\mathcal{B}}(\phi_{\mathcal{B}}(\theta), b) < \epsilon \\ \forall \theta' \in A, \theta' = \operatorname{argmax}_{\theta \in N(b_{\theta'})} f(\theta) \end{cases}$$

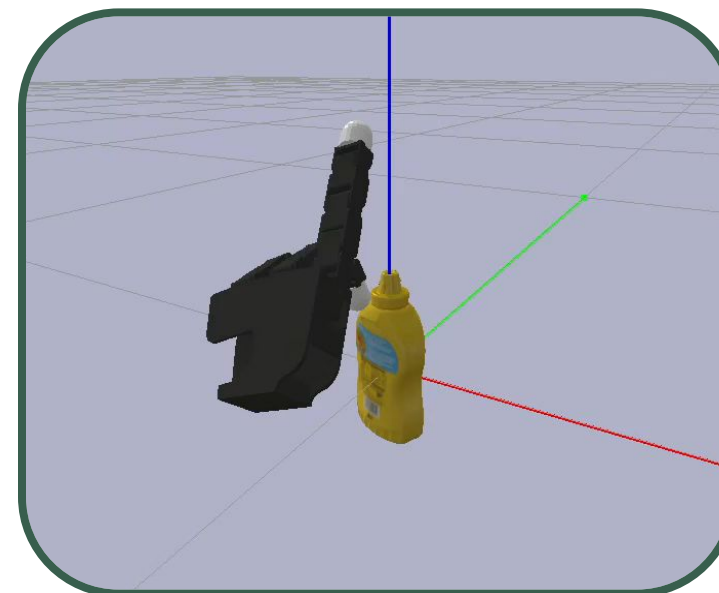
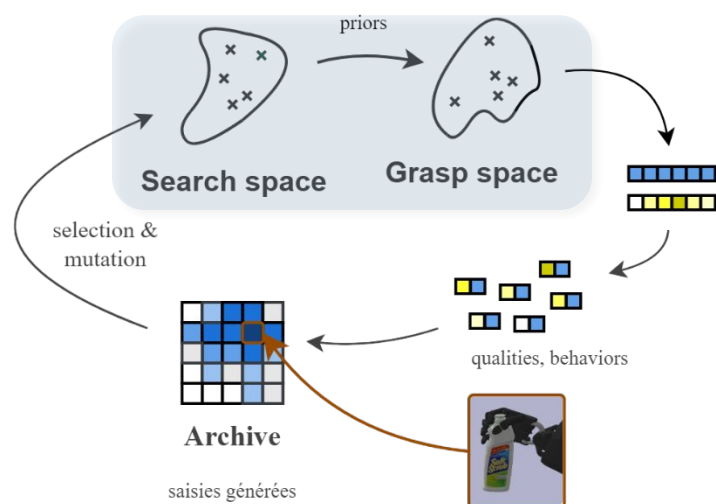
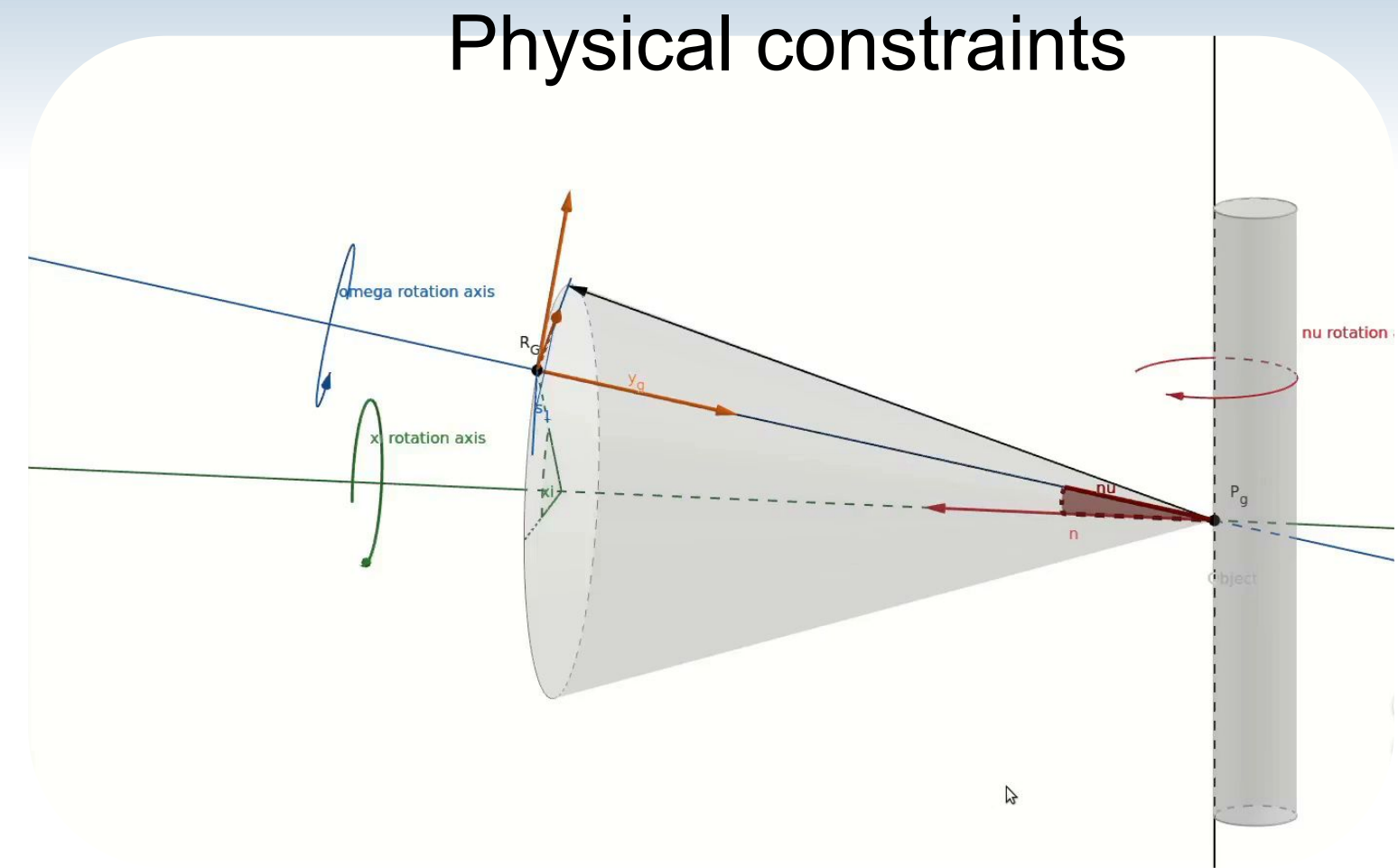
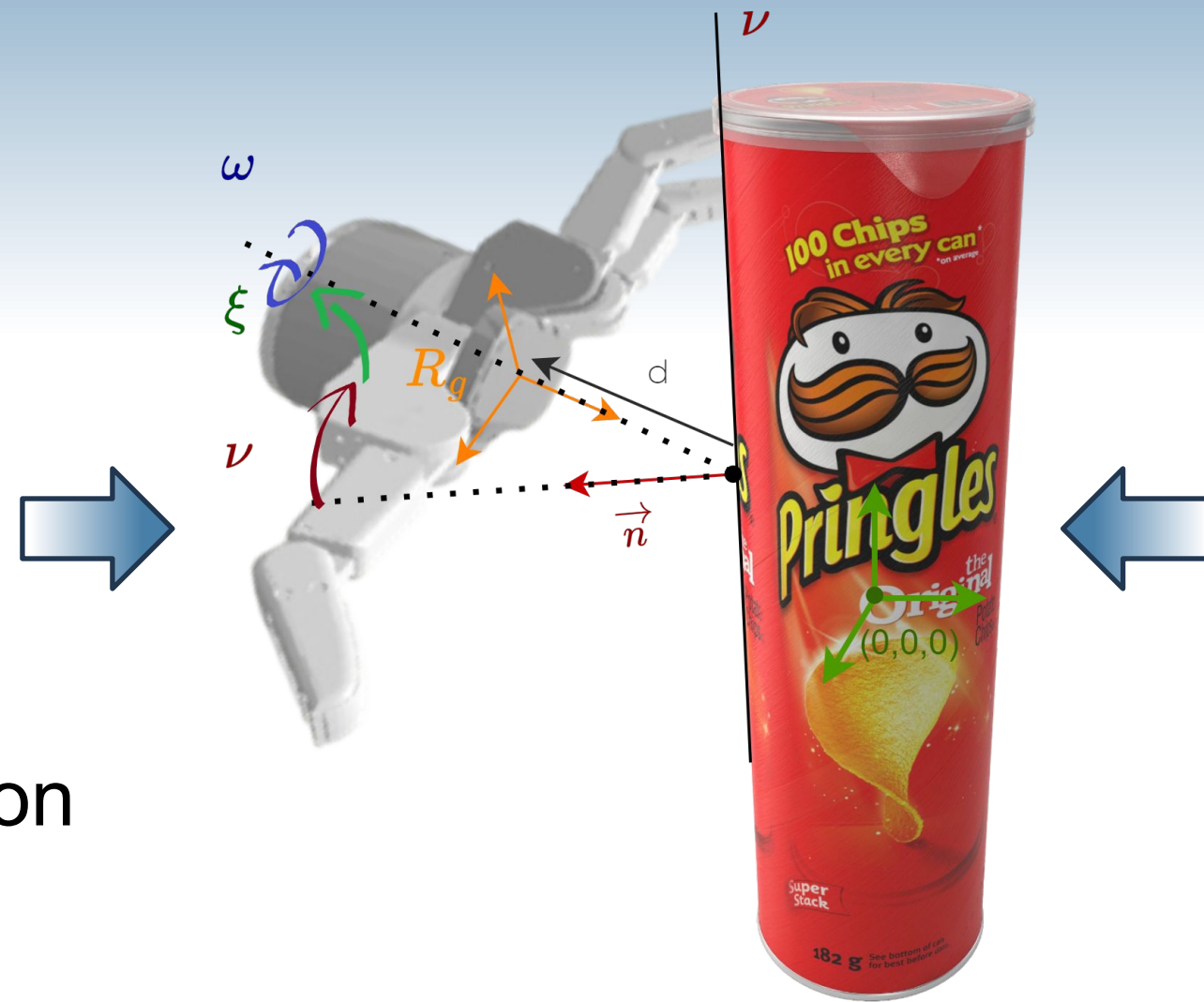
How Quality-Diversity (QD) methods can be used to speed-up 6-DoF Grasp Sampling ?

Proposed approach



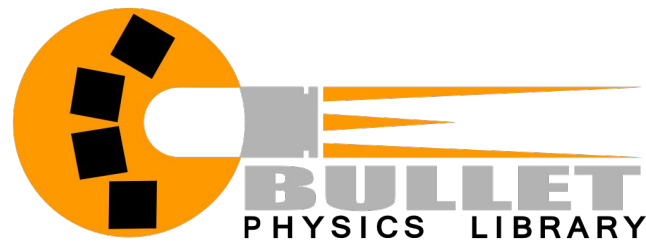
Step 1: Sample 6-DoF grasps in the search space

1 Sample
=
1 grasp candidate
=
6-DoF position
+
internal link configuration

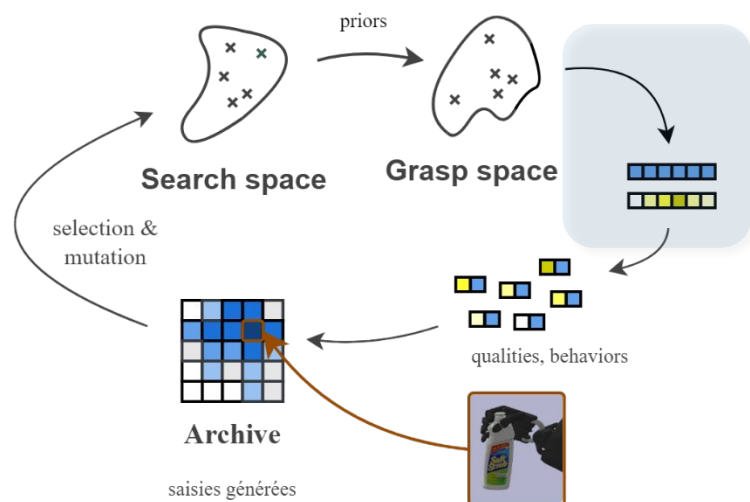
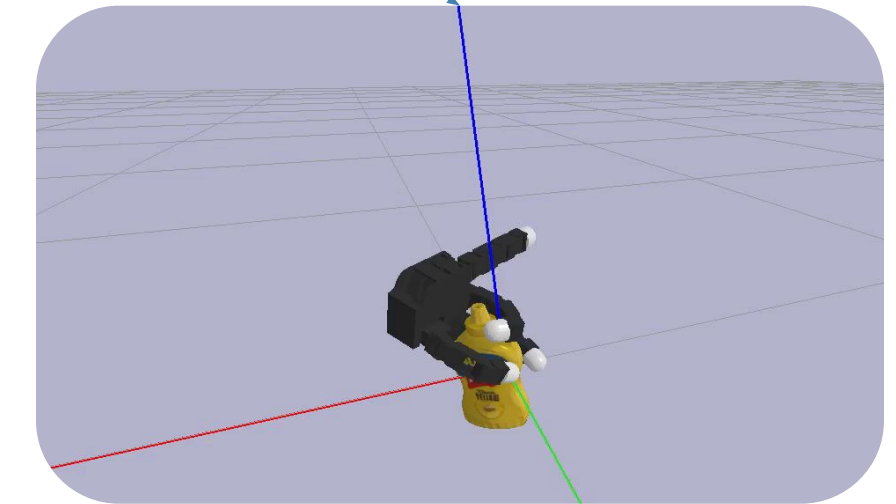
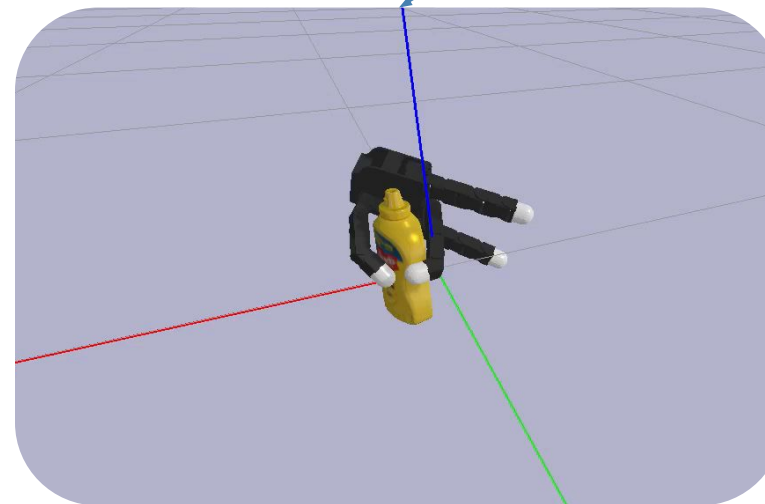


Step 2: Evaluate 6-DoF grasp samples

Physic engine to **simulate interaction**



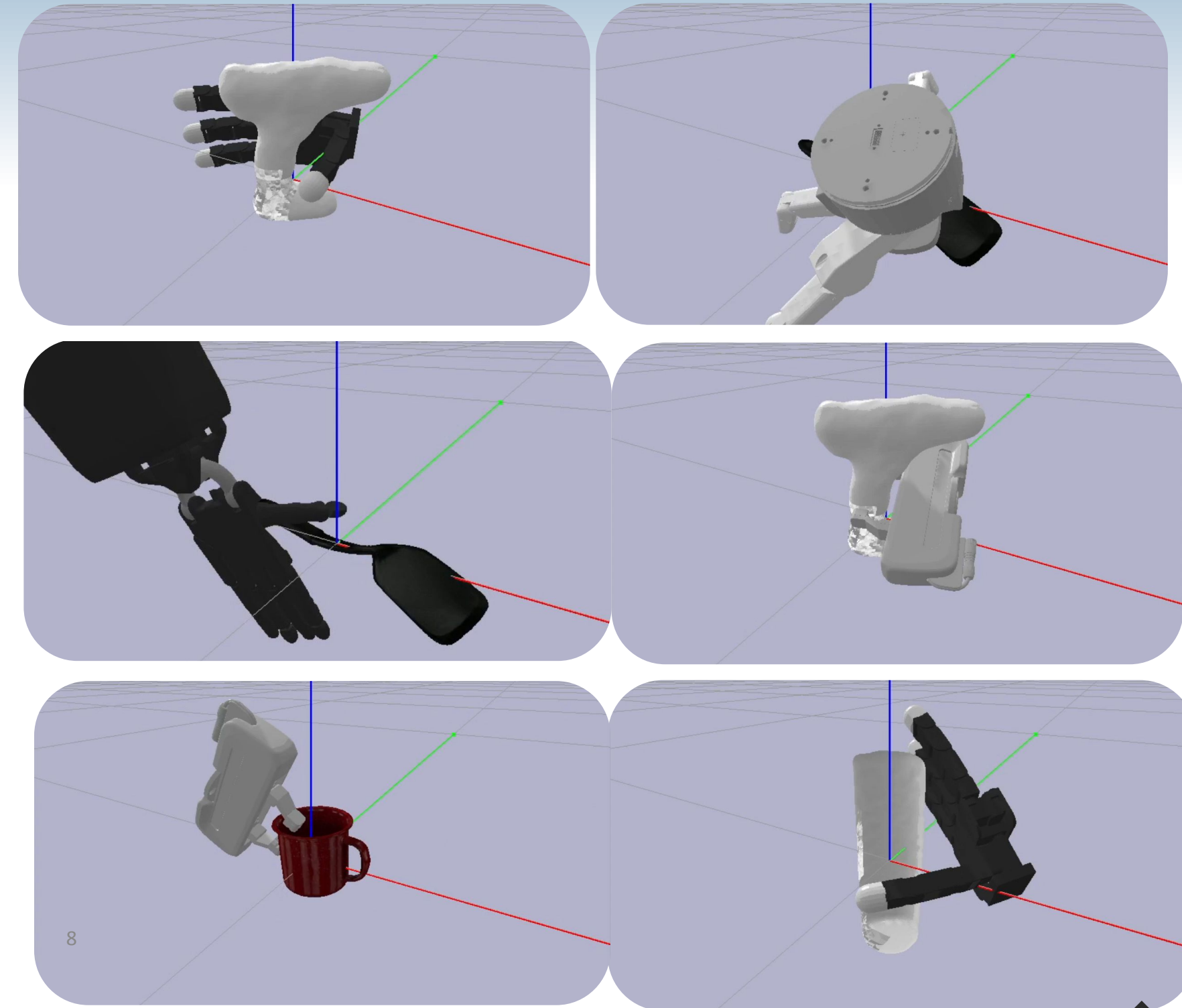
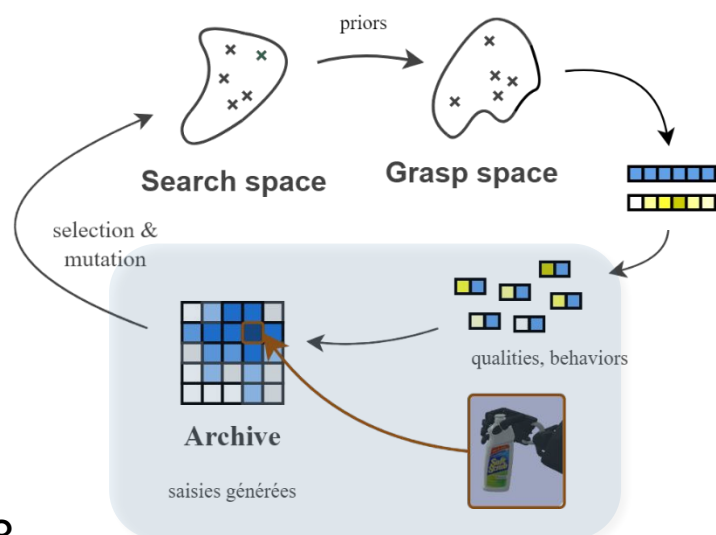
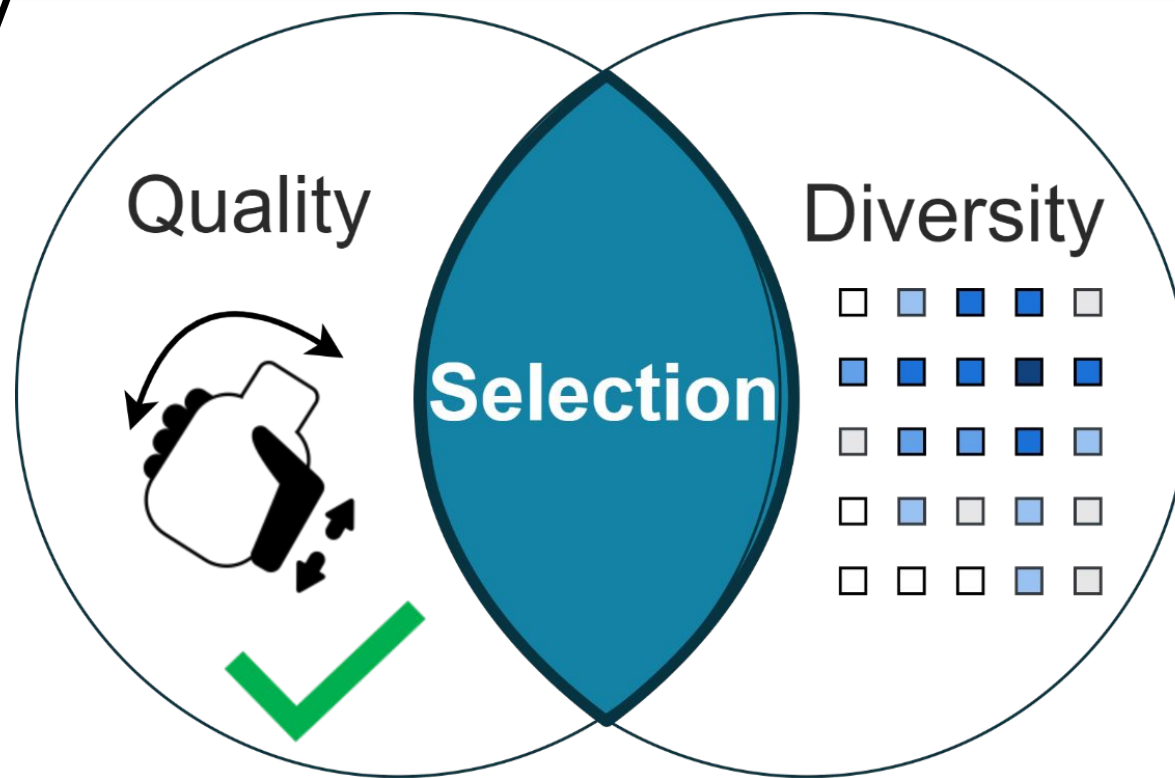
Is the grasp robust ?



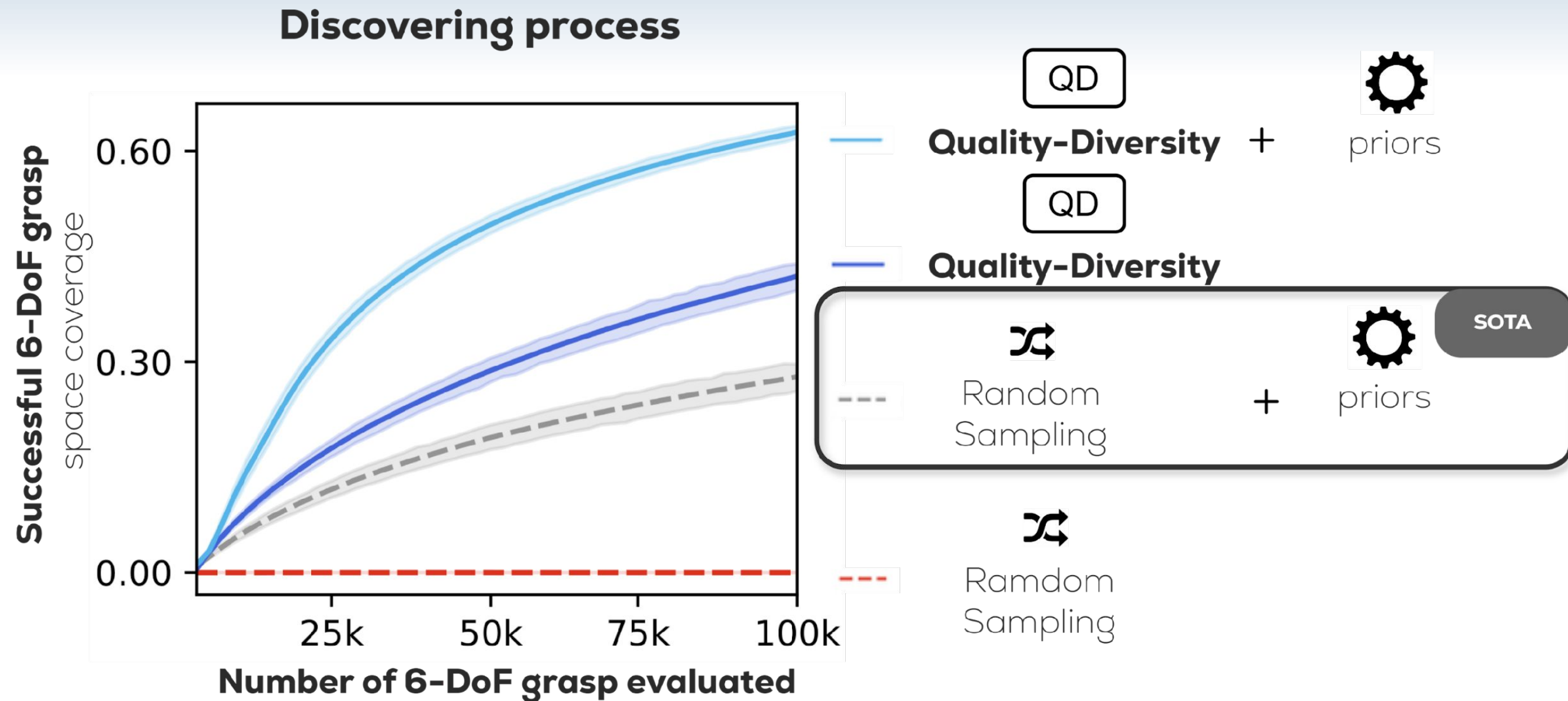
Step 3 : Archive storage and selection-mutation process

2 **Criteria** to store grasps :

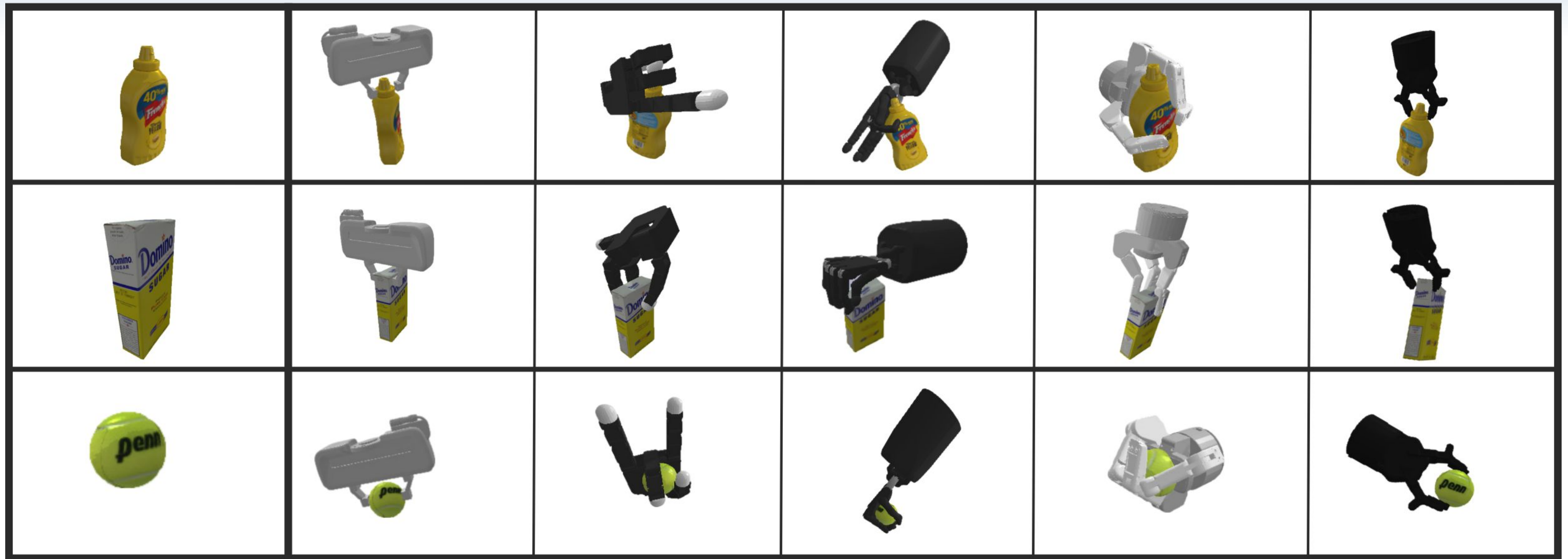
- Quality
- Diversity



Quality-Diversity methods speed-up grasp 6-DoF sampling



Quality-Diversity Methods Generate 6-DoF grasp poses



Thank you!

<https://qdgrasp.github.io/>

